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Early detection and quantification of airborne inocula of *Plasmopara viticola* causing grapevine downy mildew using impaction spore trap

Govindan Muthukumar [a], Ayyanar Kamalakannan [a,*], Irudhayasamy Johnson [a], Pachamuthu Kamaraj [b], Iyyamperumal Muthuvel [c], Shanmugam Varanavasiappan [d], Thirumalai Angayarkanni [e]

[a] Department of Plant Pathology, Tamil Nadu Agricultural University, Coimbatore, India
[b] Department of Farm Machinery and Power Engineering, Agricultural Engineering College & Research Institute, Kumulur, Trichy, India
[c] Department of Fruits, Tamil Nadu Agricultural University, Coimbatore, India
[d] Department of Plant Biotechnology, Tamil Nadu Agricultural University, Coimbatore, India
[e] Department of Biochemistry, Biotechnology and Bioinformatics, Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore, India

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ABSTRACT

Grapevine downy mildew, incited by the oomycete *Plasmopara viticola,* is an important pernicious pathogen that threatens grapevine cultivation. To facilitate early detection and quantification, this study employed SYBR Green-based quantitative Polymerase Chain Reaction (qPCR) targeting the *cellulase synthase A4* gene of *P. viticola.* Under *in-vitro* study, the pathogen was detected 4 days prior to symptom development on artificially inoculated grapevine leaves with a sporangial suspension of 1×10^4 sporangia/ml. To quantify the airborne inoculum, a standard curve was generated using tenfold dilutions (10^5 to 10^1 sporangia/ml) placed on sampling rods. On the basis of the qPCR results, a minimum inoculum threshold of 100 sporangia (Cq value = 23.98 ± 0.7) was considered for the detection of field samples. Airborne inocula of *P. viticola* were detected from the spore traps installed at heights of 0.5 m and 2.5 m from pruning to harvest and quantified using qPCR. A sampling rod collected from a height of 0.5 m was used to detect the pathogen from pruning to fruit set, with a maximum concentration of 2.8×10^5 sporangia/m^3. The assay detected the inoculum 10–14 days before visual symptom appearance in the field. Therefore, early detection of pathogen inoculum using spore trapping technology coupled with qPCR will be helpful in the development of forewarning the possible occurrence, risk prediction, and the need based fungicidal application.

1. Introduction

Grapevine Downy Mildew (GDM) incited by *Plasmopara viticola* (Berk. et Curtis) is one of the most important oomycete obligate parasites and poses challenges in the viticulture industry [1]. The pathogen mainly targets the soft, immature green tissue of leaves, twigs and fruits, resulting in rapid and significant yield losses. In the field, this infection causes devastating losses of up to 40–90 % under favorable environmental conditions [2]. *P. viticola* thrives best at 25 °C, followed by rain, which is the primary cause of downy mildew epidemics. The rapid global expansion of *P. viticola* is facilitated by its asexual reproduction cycle [3]. During the phase of intracellular mycelial growth, no visible signs of infection occur [4]. Generally, 7–10 days later, indications of initial infection emerge on the upper surface of leaves, ultimately causing yellowish oily spots, and during disease progression, these spots turn brown, ultimately leading to tissue death. *P. viticola* is a polycyclic pathogen capable of completing several infection cycles throughout a single grapevine growing season. It primarily survives in the form of oospores, which are sexual structures found in dead leaves on vineyard soil [5]. Pruning and canopy management can alter the microclimate within the grapevine canopy, thereby influencing disease development [6,7]. While some hybrids are resistant, but the majority of *Vitis vinifera* cultivars remain susceptible [8,9].

Nevertheless, the majority of the growers rely on fungicide

* Corresponding author.
E-mail addresses: muthug1999@gmail.com (G. Muthukumar), kamalakannan.a@tnau.ac.in (A. Kamalakannan), johnsonpath@gmail.com (I. Johnson), Kallaikams2000@yahoo.co.in (P. Kamaraj), im74@tnau.ac.in (I. Muthuvel), varanavasiappan.s@tnau.ac.in (S. Varanavasiappan), angait73@gmail.com (T. Angayarkanni).

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application for plant disease management. Management relies heavily on fungicide applications made at either 7–10 (wet) or 10-14-day (dry) intervals [10]. These conventional approaches might lead to the overuse of fungicides, yet alternative strategies are available to better manage their frequency of application. By delaying the initial GDM treatment, it is possible to lesser the total number of fungicide applications needed during the growing season. Some of these strategies involve monitoring critical disease levels [11,12] or tracking airborne inocula [13], whereas others are based on weather patterns to predict disease risk [14]. Weather-based predictive models, which are based on weather data, have been extensively utilized to evaluate the likelihood of plant disease outbreaks by combining environmental factors with the presence of pathogens. According to the vapour pressure deficit model, a threshold of 2.52 viable sporangia/m^3 air/day can be employed to inform farmers about the potential risk of GDM infection in vineyards with vulnerable *V. vinifera* varieties [15]. The severity of mildew epidemics is heavily influenced by several factors in vineyards such as microclimate, canopy structure, and topography [16]. Hence, weather-based disease forecasting models do not consider those factors, ultimately limiting their accuracy in disease prediction.

Alternatively, pathogen availability-based warning systems are being developed to manage crop diseases worldwide. A reduction of 3.3 fungicidal sprays was used to manage grapevine powdery mildew in Oregon State (USA) based on spore trapping technology [17]. In India, the incidence of rice false smut was correlated with the airborne inoculum of *Ustilaginoidea virens* [18].

As most sporangia are released between 8:00 p.m. and 7:00 a.m. [19], airborne inocula should be measured after 7:00 a.m. By the end of the day, farmers should receive notifications about sporangial counts, allowing them to decide on spraying. This method relies on measuring spore counts accurately and promptly. The process of identifying and counting spores predominantly relies on light microscopy, a technique that is time-consuming and requires significant expertise. The task of precisely quantifying of spores is more challenging because of the necessity of distinguishing them from other airborne particles present on the sampling surface, even when specific dyes are utilized [20]. Later, Polymerase Chain Reaction (PCR) based methods for the detection of several plant pathogens from both plant and air-biota samples were established [21–24]. However, traditional PCR techniques are unable to provide precise quantification, making them unsuitable for assessing whether airborne inoculum levels exceed action thresholds. Alternatively, quantitative PCR (qPCR) offers the advantages of simultaneous detection and quantification of target DNA with high specificity and sensitivity makes a perfect tool for timely disease management strategies.

Previously, the early detection of *P. viticola* was based on PCR assays targeting the ITS region of ribosomal DNA [25]. Nevertheless, the widespread conservation of ITS primers across fungal and oomycete species can limit their specificity, making it challenging to differentiate closely related species in complex environments. Consequently, species-specific primers that target the *Cellulose synthase gene (CesA4),* which is a crucial enzyme in the cell wall biosynthesis of Oomycetes during sporangial development and pathogenesis, have been developed [26]. Unlike the multicopy ITS region, *CesA4* is a single-copy, functional gene specific to oomycetes, which reduces the chance of cross-reactivity and improves the accuracy of quantification.

Notable achievements have been made in quantifying airborne inoculum across different pathosystems [17,27–33]. The aim of this study was to develop a rapid and sensitive method for the detection and quantification of *P. viticola* from air samples and enable the early identification of asymptomatic infections in grapevine leaf tissues.

2. Materials and methods

2.1. Collection of downy mildew infected samples

During the year 2024–25, grapevine downy mildew infected leaf samples (Fig. 1) were collected from two different locations, Royappanpatti village (9.7757°N, 77.3381°E) in the Theni district and Madampatty village (10.9698°N, 76.8598°E) in the Coimbatore district, Tamil Nadu, India, *via.* random sampling. The sites were selected based on the known occurrence of downy mildew history and prevalence [34]. Furthermore, healthy grapevine leaves were also collected from disease free fields in the TNAU orchard of the Coimbatore district (11.0069°N, 76.9309°E) in Tamil Nadu and used as a negative control. From these samples, sporangial masses were collected and stored at −20 °C for further analysis.

2.2. Genomic DNA extraction and PCR amplification

The genomic DNA was extracted from the sporangial mass of *P. viticola* infected samples following the procedure of S.L. Piccolo et al., 2012 [35]. Agarose gel electrophoresis was used to verify the genomic DNA, and the DNA concentrations were measured using a NanoDrop Lite Plus Spectrophotometer (NanoDrop products, Thermo Scientific, Wilmington, DE, USA).

The species-specific primer pair Nad9 cob-F/R, which targets the mitochondrial COB region, is a selective advantage for studying the evolutionary history of *P. viticola* [36]. On the other hand, the *cellulase synthase* gene provides high specificity because of its role in the synthesis of cellulose in oomycetes [37] (Table 1). Hence, both the primer pair COB-F/R and *Cellulase synthase A4* were used for the detection of *P. viticola* [38]. The endpoint PCR was performed using an Applied Biosystems (VeritiPro™) thermal cycler. The reaction mixture was prepared with 10 μl of TAKARA master mix (2X concentration, Catalogue no. RR310), 10 pmol forward and reverse primer (2 μl each) and 100 ng of DNA template (approximately 2 μl), with nuclease-free water added to reach a total volume of 20 μl. The PCR protocol was executed with a specific temperature parameter for the initial denaturation, annealing and extension steps mentioned in Table 1. Healthy plant DNA sample and sterile water were used as negative control in the PCR assay. To verify the amplified PCR products at 1 % agarose gel performed under a UV-transilluminator, and the products were visualized using gel documentation system (UVITEC Ltd. Cambridge, England).

2.3. qPCR assay

The qPCR assay was designed to target the outer primers of *CesA4* gene of F3:5′-CCAAGCTCTTGTCGCAGTC-3' & B3: 5′-CGACGTCGCTGTGAAAGT-3 [38] for real-time PCR based quantitative studies (Table 1). The expected DNA fragments were separated from the gel and purified using the Gene Jet gel extraction kit (Thermo Fisher Scientific, Vilnius, Lithuania). Furthermore, these fragments were cloned and inserted into T-Vector PTZ57 R/T according to the manufacturer instructions. The cloned sequences were subsequently analyzed using M13 primers at Biokart India, which is located in Bengaluru, Karnataka, India. The sequences were refined with BioEdit later BLASTn was performed, and the aligned sequences were deposited in the National Center for Biotechnology Information (NCBI) GenBank database.

To determine the optimal conditions, a range of experiments were conducted, which includes testing primer concentrations between 250 and 700 nM, as well as optimizing the annealing temperatures from 51 to 60 °C using gradient PCR. The real-time PCR mixture consisted of 5 μl of 2X TB Green® PreMix (Takara Bio Inc. Kusatsu, Japan), optimized amounts of forward and reverse primers, 1 μL of DNA template (~100 ng), and molecular biology grade water to reach a final volume of 10 μL. The Bio-Rod CFX Opus 96 Real-Time PCR System (Bio-Rad Laboratories, Kaki Bukit View, Techview, Singapore) was utilized with the following



Fig. 1. Symptom progression of grapevine downy mildew. A. Yellowish oily translucent spot appears on upper surface of the leaf; B. Correspondingly lower surface mass of white velvety mycelial growth of sporangiophore and sporangia; C. During advanced stage of infection, infected leaf turned into necrotic and premature defoliation; D. Mass of whitish fuzz sporangiophore and sporangia on young pea size berry; E. Closure view of sporangiophore; F. Berries are start approaching mummification.

Table 1
Primers used in this study.

Primer Sequence	Amplicon size	Annealing temperature	PCR cycle	References
PCR primers				
COB-F 5′-GTATAATTTATTTAAAATAAG-3′ COB-R CAAACATATCCCAAATTTC'3	520 bp	**51°C**	Initial denaturation at 94 °C for 3 min, followed by 40 cycles of denaturation (94 °C for 30 s), optimum annealing temperature, and elongation (72 °C for 90 s). A final elongation step at 72 °C for 7 min.	Basha et al. (2021)
CesA4- F ATGACCAATCGGGACGCG'3 *CesA4*- R CATGACGGTGAACTCGAG'3	530 bp	**65°C**		Marimuthu et al. (2019)
qPCR primer				
CesA4-F3CCAAGCTCTTGTCGCAGTC *CesA4*-B3CGACGTCGCTGTGAAAGT	213 bp	**58°C**	Initial denaturation at 94 °C for 5 min, followed by 40 cycles of 94 °C for 30s, optimized annealing temperature for 30s, and extension at 72 °C for 30s.	Marimuthu et al. (2019)

protocol as mentioned in Table 1. To ensure amplicon specificity, melt curve analysis was performed. This involved incrementally increasing the temperature by 0.5 °C for 5 s and the corresponding fluorescence was recorded at each step. Each sample was analyzed with three biological and three technical replicates. The replicate Cq values were used to calculate the mean Cq and standard error of the mean.

2.4. Preparation of standard curve

Absolute quantification of the plasmid DNA of *P. viticola* was achieved by the procedure described by Campos et al., 2023 [39] with minor modifications. A fragment of the *CeSA4* gene (213 bp) was eluted from the gel using a Gene Jet Kit and cloned and inserted into the T-Vector PTZ57 R/T (2887 bp) vector. Blue-white screening was employed to identify recombinants, which were then verified through colony PCR using plasmid-specific M13 primers. An EZ-10 Spin Column kit (Bio Basic Canada Inc., Markham, Canada) was used to isolate plasmid DNA, and its concentration was measured using a NanoDrop Lite Plus Spectrophotometer (NanoDrop products, Thermo Scientific, Wilmington, DE, USA) and converted to copy number according to the

formula described by Vasconcelos et al., 2021 [40].

Plasmid isolation was performed from the selected recombinants, and the DNA concentration was adjusted to 30 ng/μl. For standard curve preparation, the plasmid DNA was tenfold serially diluted seven times from 30 ng to 30 fg. This experiment was performed twice using identical DNA preparations. A negative control was included by substituting template DNA with nuclease free water. To generate the standard curve for the partial *CeSA4* gene sequence of GDM, a linear regression curve was plotted with mean Cq values on the Y-axis and the logarithmic value of serially diluted DNA (in ng) on the X-axis. Each concentration was tested with three technical replicates. The entire experiment was independently repeated two times, and nuclease-free water served as a no-template control. The overall performance of the qPCR assay was evaluated based on certain criteria, including efficiency (90–110 %), $R^2 \geq 0.98$, and repeatability (relative standard deviation $\leq$ 25 %) [41].

2.5. qPCR performance evaluation

To evaluate the specificity of the developed qPCR assay, genomic samples from 10 different plant pathogens were examined. These samples were procured from the Culture Collection Centre within the Department of Plant Pathology, TNAU, Coimbatore (Table 2). DNA was extracted from 100 mg of dried mycelial mats cultured for 14 days as per the protocol described by Murray & Thompson, 1980 [42]. The qPCR was performed as described above. Furthermore, to assess the impact of potential inhibitors on grapevine leaf tissues, DNA extracted from *P. viticola* pure genomic DNA and plant DNA was utilized. Plant DNA (50 ng) was combined with increasing concentrations of pathogen DNA (10 ng, 1 ng, 100 pg, 10 pg, and 1 pg/reaction, with three replicates each). Tenfold serial dilutions of plasmid copies (2.96×10^{10} copies/μl) of *P. viticola* were used as templates to determine the sensitivity of both the end point PCR and the qPCR assay. Additionally, a melting curve analysis was conducted to verify the amplicon uniqueness, and potential primer dimers were checked.

Table 2
Details of the fungi, oomycetes and plant used for specificity of the developed qPCR assay.

S. No.	Fungi name	Pathogenic family	NCBI Accession Number*	qPCR amplification
1.	*Phytophthora infestans*	*Pernosporaceae*	MW285753	NA
2.	*Phytophthora capsici*	*Pernosporaceae*	OR352452.1	NA
3.	*Plasmopara viticola*	*Pernosporaceae*	MN562238	21.25 ± 0.6
4.	*Plasmopara viticola*	*Pernosporaceae*	MN562239	21.38 ± 0.2
5.	*Plasmopara viticola*	*Pernosporaceae*	PV190017	21.98 ± 0.5
6.	*Erysiphe necator*	*Erysiphaceae*	MK685859.1	NA
7.	*Colletotrichum gloeosporioides*	*Glomerellaceae*	OR145043	NA
8.	*Macrophomina phaseolina*	*Botryosphaeriaceae*	MZ234156	NA
9.	*Alternaria alternata*	*Pleosporaceae*	OR244396	NA
10.	*Scelotinia sclerotiorum*	*Sclerotiniaceae*	MW151691	NA
11.	*Ustilaginoidea virens*	*Clavicipitaceae*	MZ666394.1	NA
12.	*Ampelomyces quisqualis*	*Phaeosphaeriaceae*	MW209691	NA
13.	*Pythium aphanidermatum*	*Pythiaceae*	MK758062	NA
14.	*Clonostachys rosea*	*Bionectriaceae*	MT605141	NA
15.	*Trichoderma asperellum*	*Hypocreaceae*	KX533986	NA
16.	*Vitis vinifera*	*Vitaceae*	–	NA

Abbreviation; NA- No amplification.

2.6. Detection of P. viticola on asymptomatic grapevine leaves

A time course infection experiment was to detect latent infections of *P. viticola* in grapevine leaves was conducted according to the procedure of Primiano et al., 2021 [19,43]. At BBCH (Biologische Bundesanstalt, Bundessortenamt and CHemical industry) stage 18 (approximately 20 days post pruning [44], the lower surfaces of leaves from nine potted vines were inoculated by spraying with a *P. viticola* suspension containing 1×10^4 sporangia/ml adjusted using hemocytometer. To evaluate primer specificity, three plants received a water spray as a mock inoculation. All the treatments incorporated with 0.01 % Tween 20 (HiMedia, Maharashtra, India). Following inoculation, the leaves were enclosed in damp, sealed polyethylene bags to maintain inoculum droplet moisture. These bags were removed after 12 h. At 2, 5, and 7 days post-inoculation (dpi), three leaf discs (2 cm in diameter) were excised from the third leaf. Leaf discs from mock-inoculated plants were also collected. Each disc was obtained from a distinct plant, and served as a biological replicate, resulting in three biological replicates for each time point.

2.7. Devising low-cost impaction spore traps

A solar power-operated impaction spore trap was developed by following the design provided by Mahaffee, 2014 [16] with slight modifications (Fig. 7). A low-cost solar-power operated impaction spore trap was designed with the help of a miniature DC motor, a Teflon disk (6.5 mm) with a 1.5 mm hole in the center and vacuum grease placed underneath, an aluminum sampling arm with a sampling radius and stainless-steel sampling rods. The DC motor with a speed of 500 rpm is operated by a solar rechargeable 5 mAh sealed lead battery. The battery is recharged by 12 V solar panels attached with a height adjustable stand. The spore trap was capable of processing 48.3 ± 1.3 L air/min by spinning two stainless steel rods at 1.151 m/s. The height of the spore trap stand is adjustable so that the spore trap as per the growth stages of crop. The sampling rod was prepared according to the protocol of Arumugam et al., 2024 [45].

2.8. Detection and quantification of airborne inocula of grapevine downy mildew using impaction spore traps

Two custom impaction spore traps were installed at a height of 0.5 m and 2.5 m from ground level in a grapevine field located in Onapalayam village of Coimbatore district, India (Latitude 11.0102° N, Longitude 76.8671° E) (Supplementary Fig. 1). Both spore traps were operated continuously from pruning to the harvesting stage of the grapevine crop from the 50th to 13th standard week of Rabi, 2024-25. The sampling rods were removed from the spore traps on a weekly basis and replaced with one another. DNA was extracted from the sampling rods according to the protocol of Basha et al., 2021 [23]. The extracted DNA was stored at −20 °C for further studies.

To quantify the airborne inocula of *P. viticola,* a standard curve was prepared using tenfold sporangial dilutions (ranging from 1×10^5 to 1×10^1 sporangia/ml) placed on stainless steel sampling rods. The DNA was extracted from the sampling rods according to the protocol of Arumugam et al., 2024 [45]. This process was repeated three times to generate an independent *P. viticola* sporangial dilution series. The final standard curve was prepared by calculating the average Cq values for each sporangial quantity from the five independent DNA extractions. Each standard curve was repeated once. Based on the extensive analysis of known quantities of sporangial suspensions, we were able to interpolate the approximate levels of airborne sporangia collected within 7-day sample collection intervals.

After optimization of the standard curve, a qPCR assay was

performed for the absolute quantification of airborne inocula of the GDM pathogen at weekly intervals. All the qPCR assay were performed according to the above protocol, and every reaction contains 1000 sporangia as a positive control. Sporangial quantification was determined for each unknown air sample by identifying the average Cq value for each triplicate reaction at which the log-linear phase intercepted the 0.05 threshold value and comparing this value to a known standard curve.

2.9. Statistical analysis

The CFX Maestro software version 2.3 automatically determined the threshold value for differentiating positive and negative qPCR results and generated the standard curve. The prepared standard curve was utilized for absolute quantification to estimate airborne inoculum density. Additional data analysis was performed using Microsoft Excel. All the experiments were done with three biological replicates. For the time course infection experiments at 2,5 and 7 dpi, water was used as the control. The data are expressed as the mean ± standard error (SE). Statistical significance was tested using analysis of variance, and Bartlett's test was subsequently used to check the homogeneity of variances [46]. Differences were regarded as significant at $p < 0.001$. To evaluate the spore count differences at 0.5 and 2.5 m spore trap heights, a Wilcoxon signed-rank test was performed. All statistical analyses were performed using RStudio v. February 1, 5033 [47].

3. Results

3.1. Symptom identification and molecular confirmation

Under natural field conditions, infected samples were taken from leaves, shoots, and clusters. The first signs of the disease were oil-spot lesions on the upper surface of the leaf corresponding to the lower surface, which were white velvety mycelial growth underneath, which eventually resulted in necrosis and defoliation. Under ideal conditions, sporangia cover young berries, and the berries eventually become mummified (Fig. 1).

The DNA extracted from the downy mildew-infected leaf samples was amplified through the specific primer pair Nad9 cob-F and Nad9 cob-R**.** Later, 1 % agarose gel electrophoresis revealed an amplicon size of 520 bp, as shown in Fig. 2, and the products were sequenced. The sequence results were analyzed through BLASTn and the results revealed that the studied strains shared 99.42 % identity with GenBank accession GQ258976.1, which belongs to the *P. viticola* isolate and was submitted to the NCBI GenBank (accession number: PV190017). Healthy and nuclease-free water showed no amplification.

3.2. Optimization of the qPCR assay

The genomic DNA of downy mildew was amplified using F3 and B3 primers, which target the *CesA4* gene through gradient PCR. To confirm successful amplification, the PCR products were subjected to 1 % agarose gel electrophoresis. The gel analysis revealed a distinct amplicon of ~213 bp (Fig. 3).

Optimization experiments revealed that 58 °C and 400 nM were the best annealing temperature and primer concentration respectively, for the qPCR experiment (Fig. 3 & Table 3). However, the expected amplicon was observed between 55 and 57 °C, but a clear-cut band and an absence of contamination were observed at 58 °C using gradient PCR. This temperature was further utilized for qPCR experiments.

3.3. Cloning and standard curve preparation

A portion of 213 bp *P. viticola* DNA fragment was eluted and cloned into the PTZ57 R/T (2887 bp) vector. The recombinant plasmid sequence was authenticated through sequencing and BLASTn homology analysis, confirming that the amplification was specific to *CesA4* of *P. viticola*. The sequence results revealed that studied strains shared 100 % identity with the GenBank accession MN509558, which belongs to the *CesA4* of *P. viticola* isolate. The sequences were submitted to the NCBI GenBank (accession number: PV299257).

The standard curve was generated by plotting tenfold serial dilutions of cloned *CesA4* DNA ranging from 30 ng to 30 fg. The resulting standard curve revealed a strong linear relationship between the log of plasmid DNA copies and Cq values, resulting in a high ($R^2 = 0.994$) correlation coefficient with relatively high amplification efficiency (E = 99.2 %). Ultimately, these results will generate a regression equation of $Y = -3.341X+40.553$. This method could be utilized for the early detection and quantification of *P. viticola* infection in latent infection experiments (Fig. 4).

3.4. qPCR performance testing

To evaluate the *P. viticola* isolates and nontarget DNA samples from the 10 different plant pathogenic fungi listed in Table 2, primer specificity was tested. The developed qPCR primers successfully amplified *P. viticola* samples, as indicated by a clear amplification curve, whereas no amplification was observed for nontargeted pathogens, confirming the high specificity of the designed primers. These results confirm the high specificity of the utilized qPCR primers, which target the *CesA4* gene of *P. viticola* for real-time quantification studies.

In the assay, a ten-fold dilution of *P. viticola* DNA (10 ng–1 pg) was spiked with *V. vinifera* genomic DNA (50 ng/μl) to assess the sensitivity

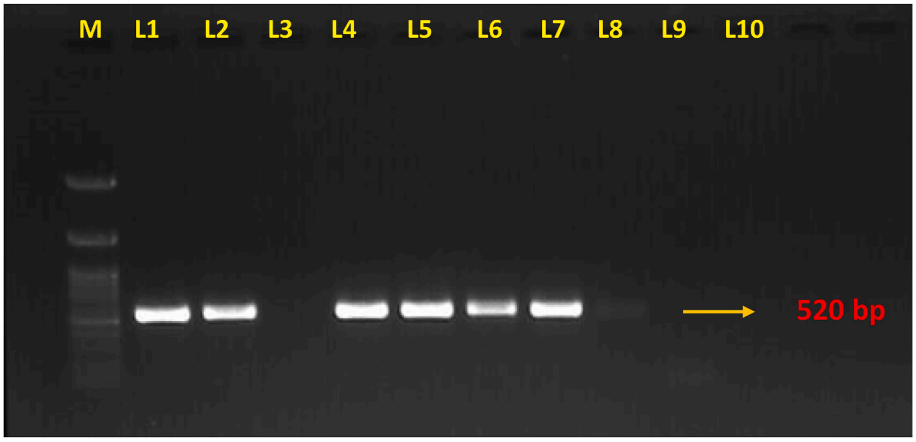


Fig. 2. PCR amplification with apocytochrome (COB) region of *Plasmopara viticola* isolates. L1-L5: Royappanpatty isolates: L6-L9: Madampatty isolates; L10-Negative control. M – 100 bp ladder.

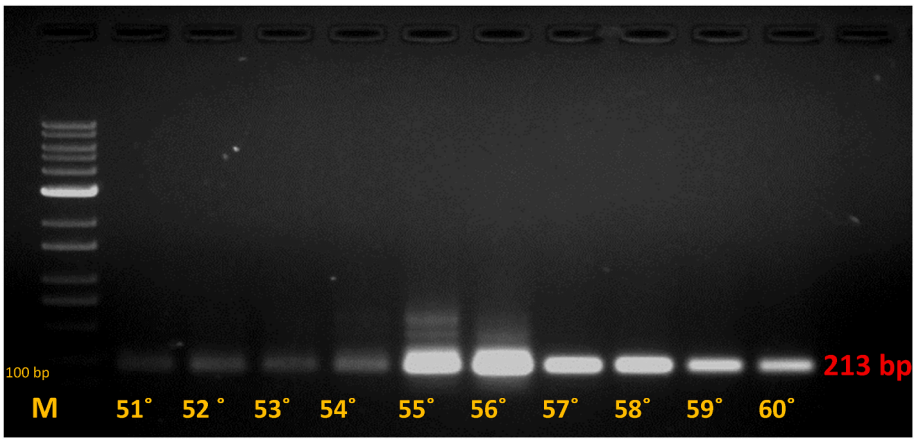


Fig. 3. Gradient PCR for optimization of annealing temperature from 51 to 60 °C for the quantitative PCR assay.

Table 3
Comparison between the amplification of *Plasmopara viticola* DNA spiked with *Vitis vinefera* DNA.

P. viticola DNA (ng/μl)	*Vitis vinefera* DNA (ng/μl)	Cq value (mean ± SD)
10	–	12.35 ± 0.46
	50	12.42 ± 0.31
1	–	15.20 ± 0.22
	50	15.45 ± 0.24
0.01	–	18.38 ± 0.36
	50	18.80 ± 0.12
0.001	–	21.21 ± 0.01
	50	21.90 ± 0.19
NTC	–	–
	50	–

Abbreviation- NTC-No template control.

of the developed assay. The results were consistent with those obtained from pure *P. viticola* genomic DNA, without limiting the amplification efficiency or sensitivity of the developed qPCR assay (Table 3.) (Supplementary Fig. 2).

Additionally, melting curve analysis was performed to determine the presence of nonspecific and potential contamination by observing multiple melting peaks. A single dissociation peak was observed at 87 °C, confirming the absence of above-mentioned contamination (Supplementary Fig. 3). The results obtained from tenfold dilutions of *P. viticola* plasmid copies were used as a template, starting from a concentration of 30 ng/μl each (2.96×10^{10} copies/μl) to assess the sensitivity between end-point PCR and the qPCR assay. As reported in (Table 4 and Supplementary Fig. 4), real-time qPCR was able to detect plasmid DNA concentrations of up to 30 fg, while the end-point PCR assay yielded 300 pg.

3.5. Detection of *P. viticola* on asymptomatic grapevine leaves

Based on the standard curve, the number of DNA copies of the pathogen were calculated for each time point. A latent infection experiment confirmed that *P. viticola* could be detected as early as 2 dpi (Cq = 25.87 ± 0.5 equal to 7.98×10^3 DNA copies/ml) (Fig. 6). This early detection clearly indicates the advanced colonization of the pathogen during the asymptomatic stage, demonstrating the high sensitivity of the qPCR assay. As number of days increased from 5 to 7 dpi, the number of pathogenic DNA are also increased from 4.19×10^4 to 5.58×10^6 DNA copies/ml, with Cq values of 23.98 ± 0.5 and 19.97 ± 0.8 respectively. No amplification (N/A) or melting curve was observed in leaves sprayed with only water. Hence, this assay detected the asymptomatic colonization of *P. viticola* up to 5 dpi.

3.6. Detection and quantification of airborne inocula of grapevine downy mildew using impaction spore traps

The standard curve generated for the *P. viticola* sporangial suspension placed on the sampling rods showed an average Cq value of 12.85 ± 0.5, corresponding to 10^5 sporangia/ml (Fig. 5). The resulting standard curves shows an (R^2 = 99.70 %) regression equation of Y = −3.40x+41.122. As little as 10 sporangia (Cq = 29.02 ± 0.4) were detected *via* qPCR assay with an amplification efficiency of 107 % considered to be qualitatively confirm the presence of infection. For the rods, a 100 % detection rate was consistently observed for 100 sporangia/rod. Based on the results, a minimum inoculum threshold equivalent of 100 sporangia was considered for initiating fungicide application using spore trapping technology. All the positive results below this threshold qualitatively confirmed the presence of sporangia.

The custom impaction spore sampler was capable of processing 487.86 L of air over a seven-day period. In the qPCR assay, air samples collected from 0.5 to 2.5 m installed spore traps revealed notable temporal variations in sporangial counts from December 2024 to January 2025. The assay detected the inoculum 10–14 days before visual symptom appearance in the field. A sampling rod collected from a height of 0.5 m was used to detect the pathogen from pruning to fruit set, with a maximum concentration of 2.8×10^5 sporangia/m^3. During the three-leaf stage, the spore count at 0.5 m reached a Log_{10} value of 8.14, which was nearly double the log unit of 2.5 m spore trap count on the same week. Thus, observing airborne inoculum density close to the canopy level (0.5 m) might be a reliable early warning method for downy mildew infection in grape vineyards (Table 5 & Fig. 8).

The Wilcoxon signed-rank test was conducted to compare the log-transformed spore densities in the 0.5 m and 2.5 m spore traps. The results revealed a significant difference in spore density between the two heights, with W = 3, Z = −2.31, and $p < 0.01$. The results revealed that the spore density was significantly greater at 0.5 m than at 2.5 m, with a mean rank difference of 1.49.

4. Discussion

Plasmopara viticola, a highly destructive pathogen, severely affects yield and causes considerable economic loss. Previously, the quantification of *P. viticola* was estimated from different sources, such as direct extraction from infected diseased and senescent leaves [25,48], Oospores in soil, leaf residues and asymptomatic leaves [49] by targeting the internal transcribed spacer (ITS) region. In the present study, SYBR-Green based detection and quantification of *P. viticola* based on the *CesA4* gene was established with high accuracy and specificity over previously developed ITS-based quantification method. SYBR-Green technology is more preferable than Taq-Man probe for testing large

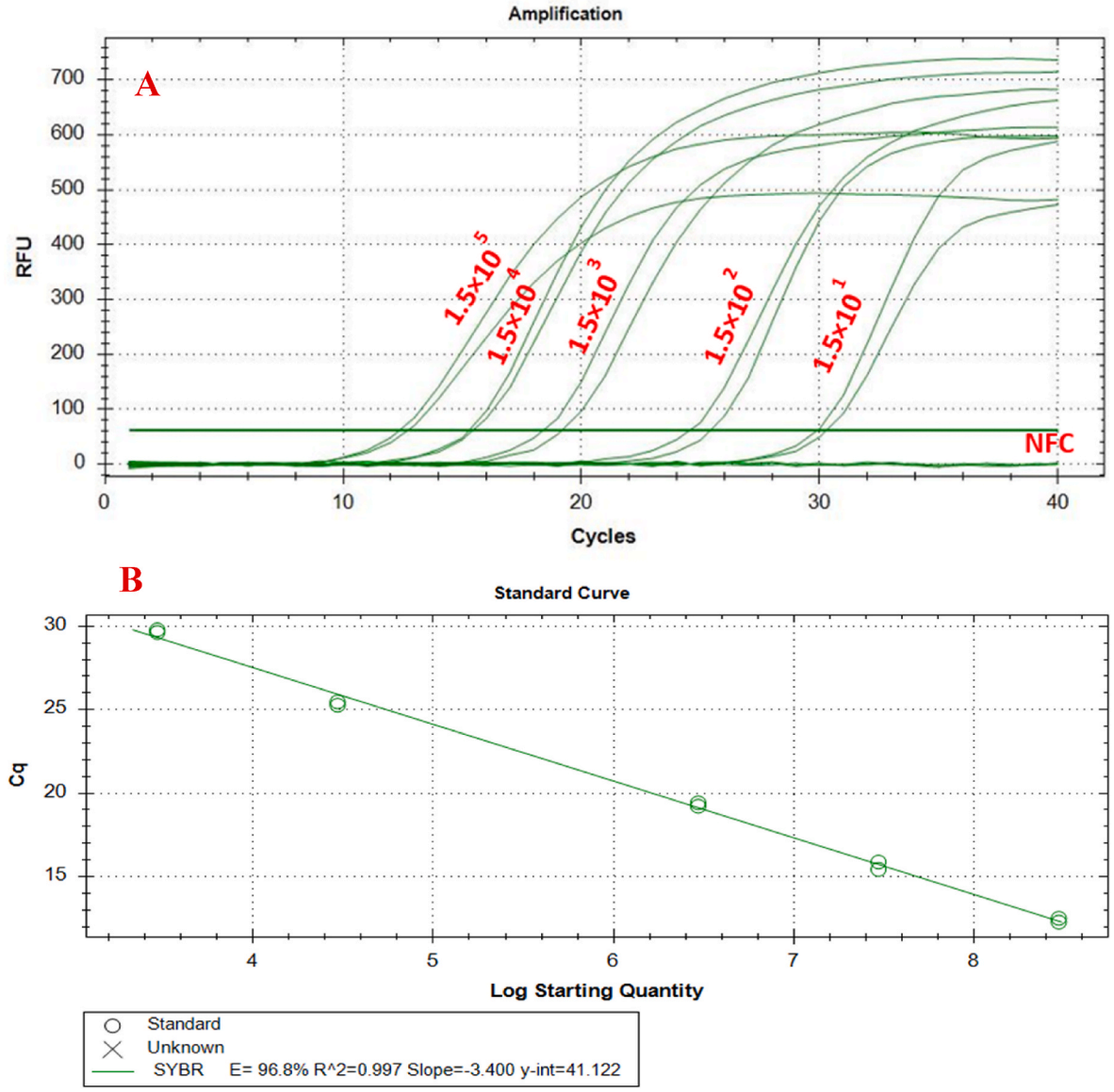


Fig. 4. Amplification & standard curve of sporangial suspension placed on sampling rods (**A & B)** showed a ten-fold dilution of sporangial suspensions (10^5 to 10^1 sporangia/ml) and its respective standard curve and related statistics.

number of samples, due to its cost-effective nature and reduced complexity. Analysis of the *CesA* gene sequences uncovered more than 100 single nucleotide polymorphisms (SNPs), resulting in a nucleotide diversity of 0.012/kilo base. Owing to this nucleotide diversity, the genome of *P. viticola* might exhibit nearly double the diversity present in the *Arabidopsis thaliana* genome [50,51]. The markers based on single copy nuclear DNA in qPCR provided high quantification potential of *Pseudopernospora cubensis* inocula than mtDNA markers [52]. Therefore, leveraging the high nucleotide diversity of the *CesA* gene has more advantages than ITS based multi-locus gene.

Notably, spiking *V. vinifera DNA* into the pathogen DNA resulted in consistent amplification without compromising the efficiency of the qPCR assay. Hence, the developed assay depicts the robustness in detection and useful for quantification of *P. viticola* infected field samples. Turco et al., 2021, reported that, chestnut plant tissues spiked with *Gnomoniopsis castaneae*, the pathogen was detected consistently without interference from plant DNA and other inhibitors present in the reaction mixture [53]. On the other hand, integrating pistachio DNA into *Septoria pistacearum* resulted in a tenfold lower limit of detection (LOD) than pure pathogenic DNA [54].

qPCR has been explored for the measurement of asymptomatic infections caused by pathogens in several instances. This technique not only saves time but also provides enhanced sensitivity and real-time quantitative detection of infected samples [55]. The high sensitivity of the qPCR assay enables researchers to measure pathogen levels in host tissues during the initial stages of colonization prior to the symptom appearance [56–58]. In our study, quantitative detection of *P. viticola* DNA in sporangial suspension inoculated plants revealed an asymptomatic appearance despite increasing fungal biomass within a limited period of 2–5 dpi. At 7 dpi, the plant exhibited a pale yellowish oily spot-on upper surface of the leaf corresponding to the lower surface mass of the fuzz sporangiophore, and sporangia were observed. According to Leroy et al., 2013, during the period of intercellular mycelial growth, infected leaves remain asymptomatic [4]. During latent infection, *P. viticola* produces early infection structures such as substomatal vesicles and primary hyphae at 24 and 36 h post inoculation (hpi) respectively, indicating early colonization of the pathogen within the host [39]. Thus, our findings directly correlate with this early intercellular colonization stage. Depending on the favorable environmental condition, the disease take leads and completes the life cycle from 5 to 18 days

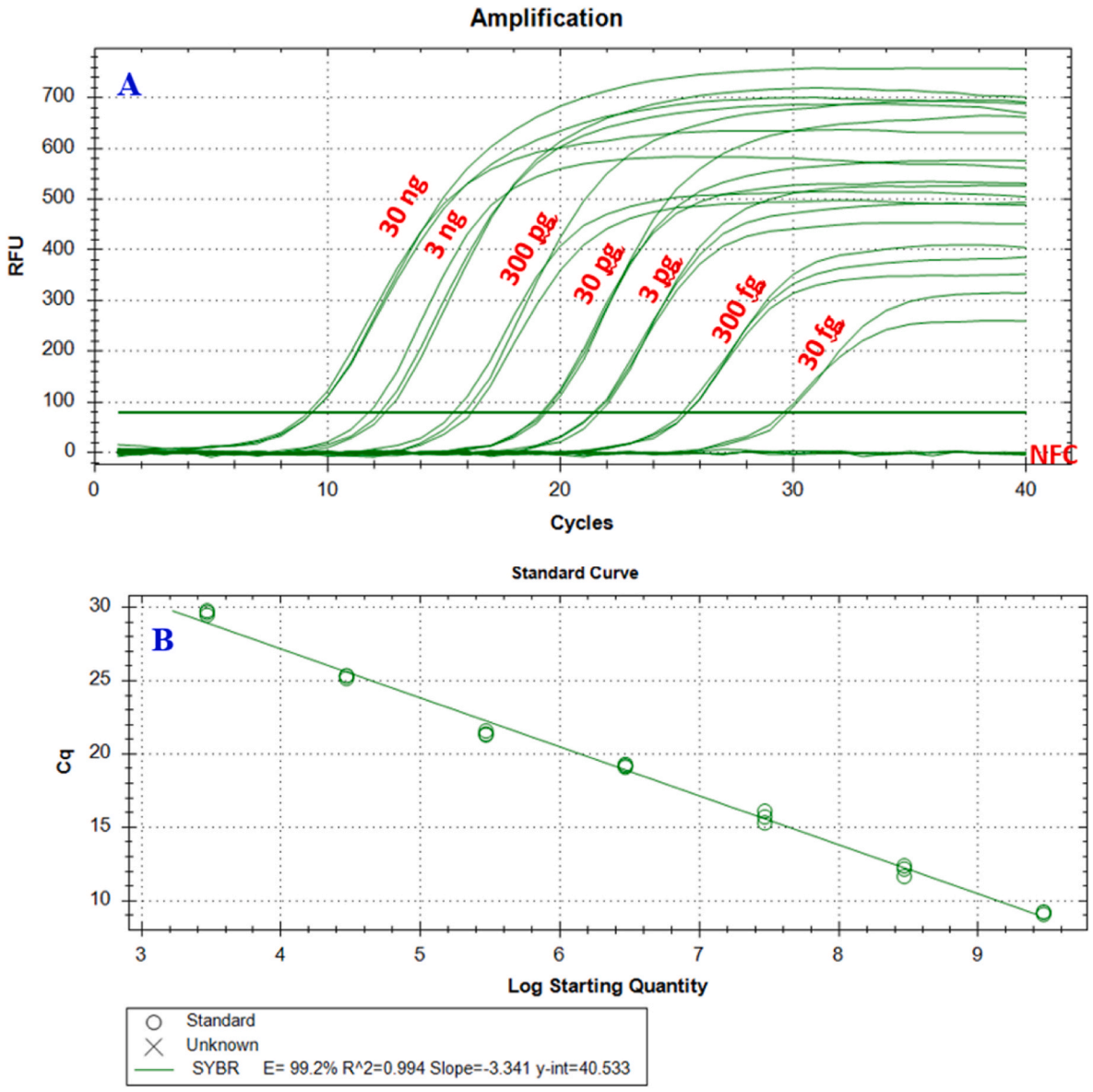


Fig. 5. Amplification and standard curve of *Plasmopara viticola* plasmid DNA (**A & B)** showed a ten-fold dilution of *P. viticola* plasmid copies of 30 ng (2.96×10^{10} copies/μl) to 30 fg/μl) and its respective standard curve and related statistics.

Table 4
Comparison of end-point PCR and qPCR sensitivity.

Starting plasmid DNA concentration (30ng/μl) (2.96×10^{10} copies/μl)								
Type of assay	10^{0}	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	NFC
End-point PCR	+	+	+	+	+	–	–	–
qPCR	8.32 ± 0.6	11.73 ± 0.2	13.86 ± 0.9	16.65 ± 0.5	21.98 ± 0.3	24.98 ± 0.6	28.90 ± 0.4	–

[59]. Therefore, the assay developed in this study can be helpful in breeding programs to assess the quantitative resistance of the grapevine cultivar to *P. viticola.*

Previously, Brischetto et al., 2021 noted that a vapour pressure deficit model was used to predict the infection rate of airborne sporangia of *P. viticola* [15]. However, this approach does not integrate a highly sensitive qPCR assay for quantifying *P. viticola* DNA, thereby limiting its ability to validate the infection threshold by molecular assays. Additionally, spore viability-based models provide only epidemiological insights but does not confirm the pathogen presence with high specificity. To address this limitation, a qPCR assay developed in this study used to validate the minimum detection threshold of the sporangial concentrations from the sampling rod. Research has shown that a qPCR assay was successfully detected the minimum inoculum threshold of 25 spores of *Nothopassalora personata* for detection of field samples [60]. Hence, overall, this approach enhances the accuracy of disease prediction compared with the viability-based approach and offers a more reliable method for early detection and risk assessment in grapevine vineyards. Similarly, in other pathosystems, 10 sporangia/ml for *P. cubensis* and *P. humili* [61], 8.548 sporangia/m^3 air for *Bremia lactucae* [62], 10 sporangia/g of dry soil for *Pernospora destructor* [63] were developed and reported previously.

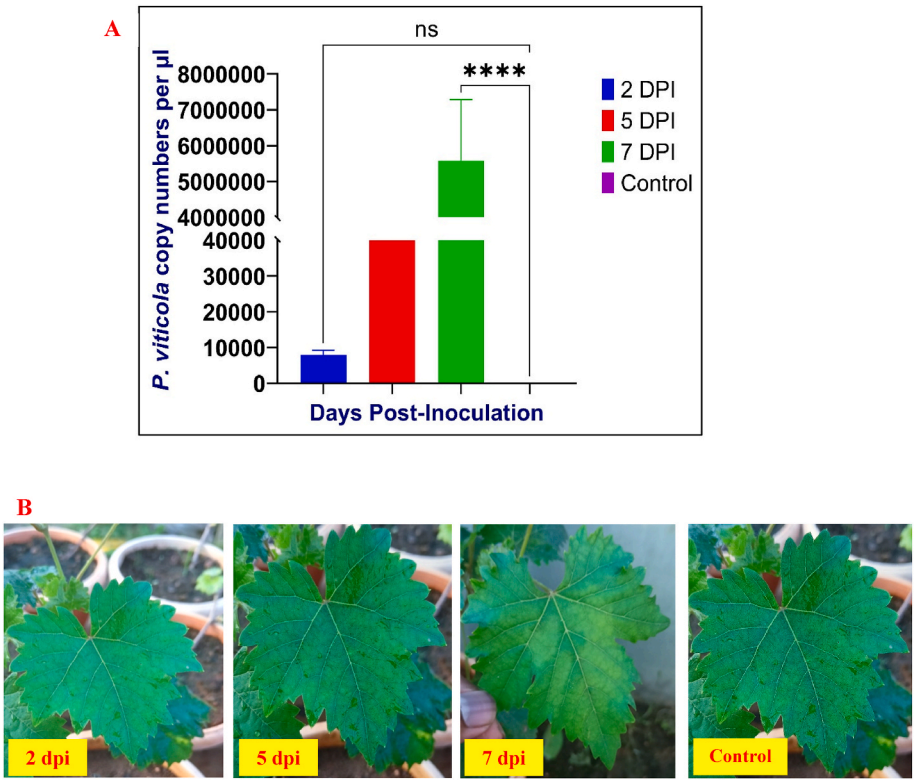


Fig. 6. Time course infection experiments to detect latent infections of *Plasmopara viticola* on grapevine leaves at 2, 5 and 7 days post inoculation (dpi). Water served as control (**A & B**). Each bar represents the mean of three individual replicates ± standard error and Homogeneity of variances was confirmed using Bartlett's test, and one-way ANOVA was applied to assess statistical significance ($p < 0.001$).

In the downy mildew pathosystem, a majority of the studies have focused on the horizontal placement of traps and the number of traps required for effective trapping of airborne inocula for timely management strategies [64–66]. In this study, the impaction spore trap installed at a height of 0.5 m showed a higher sporangial count than did the 2.5 m installed trap. A sharp decline in the spore count from the three-leaf to inflorescence emergence stage might be due to reduced sporulation or a reduction of inoculum source. This variation suggests that more inoculum is present near the ground level because of the close proximity to the infection source and the reduced vertical dispersion of the inoculum. In contrast, the apple scab pathosystem increased the number of ascospores from 0.3 to 3 m above ground level due to the effect of wind [32]. However, to better comprehend infections in the crowns of tall trees, it is essential to assess spore densities at heights exceeding 3 m and in areas with different levels of infection [67]. This insight will be helpful for the exact deployment of spore traps and will support timely decision strategies. The continuous application of chemical fungicides has also resulted in the development of resistant races of mildew causing pathogens, a reduction in the quality of fruits and a corresponding management strategy [68]. In India, mostly grapevine mildews are managed after the establishment of mildew symptoms on leaves and fruits. Since downy mildew is airborne, early detection of pathogens in the air and subsequent need based fungicidal sprays will reduce the number of sprays and increase the quality of fruits. Hence, there needs a research work on spore trap based advisory for fungicide application to reduce the indiscriminate use of fungicides. A similar approach was successfully implemented, and 3.3 number of fungicidal sprays were reduced in Oregon state (USA) based on airborne inocula availability using spore trapping technology to manage grapevine powdery mildew [17].

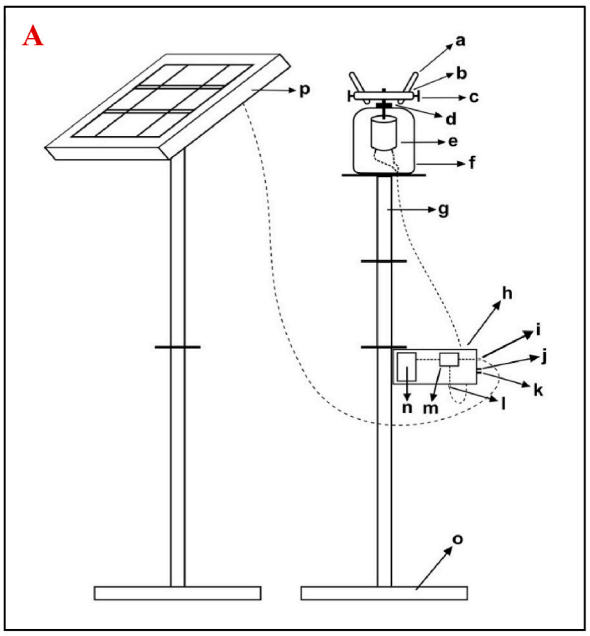
A higher number of sporangia were detected in spore traps placed at lower heights, but few environmental factors may influence the spore trapping efficiency. Research has shown that wind dynamics play a primary role in the movement of sporangia, as higher wind speeds and directionality can facilitate the upward transport of sporangia above the canopy, ultimately rendering traps placed at lower heights less effective [66,69]. Secondly, physical barriers such as dense foliage can filter or obstruct vertical air movement, thereby limiting the upward dispersal of spores from lower strata [17,70]. Temporal and spatial variations in vineyard topography change over time, which can influence the localized air currents or spore movement effects that can also interfere with accurate detection.

A key limitation of spore trap deployment in commercial vineyards is that small sampling surfaces can quickly become overloaded in dusty environments [13]. Weather conditions like prolonged cloudy days, may also reduce the effectiveness of solar-powered spore traps. Additional limitations of the present study include the need for regular maintenance, the frequency of sampling, and subsequent molecular analysis. Future research should focus on integrating spore traps with molecular biosensors for real-time detection, which would ultimately reduce the reliance on manual labor. Standardizing the number of spore traps per vineyard could ensure the precise monitoring of airborne sporangia across different vineyard settings, thereby enhancing the reliability of spore trap-based advisories.

To the best of our knowledge, this is the first report of SYBR-Green based real-time quantitative detection of the *P. viticola* pathogen targets *CesA4* gene using impaction spore trap. Based on the present study, a 0.5 m installed spore trap could serve as an effective height for early warning systems of downy mildew outbreaks in grape vineyards.

5. Conclusion

In this study, a *CesA4* gene-based qPCR assay demonstrated high sensitivity for the asymptomatic detection and quantification of *P. viticola* sporangia in grapevine leaves and spore trap samples respectively. The developed assay exhibited the lowest detection limit, was capable of detecting genomic DNA at the femtogram level, and



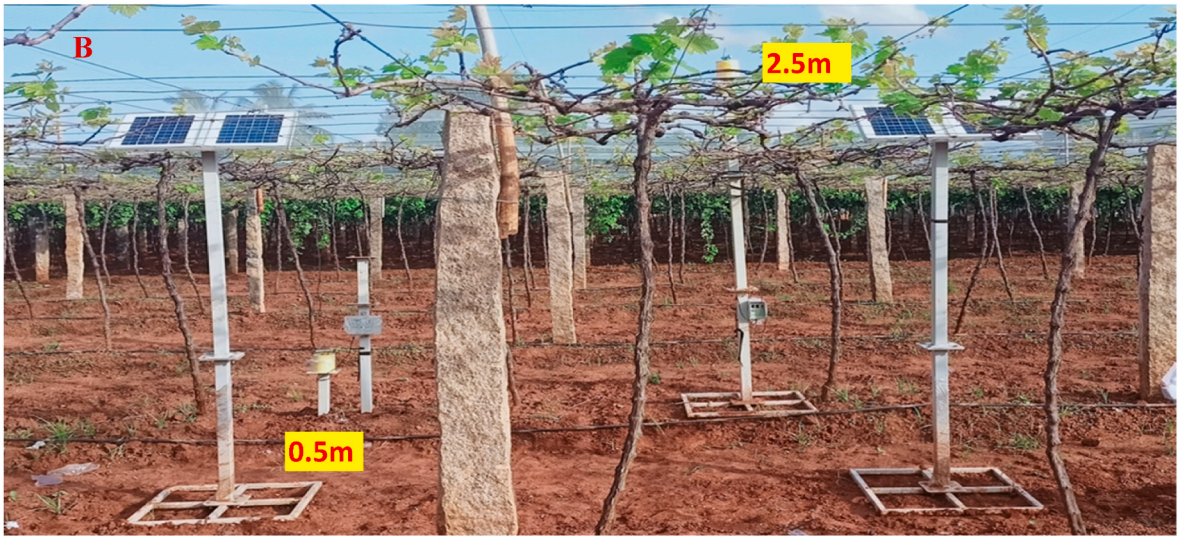


Fig. 7. Low-cost impaction spore trap. Lay out of solar operated impaction spore trap and its components for the detection of airborne inocula of grapevine downy mildew pathogen **(A). a)** Sampling metallic rods (50 × 7 mm); **b)** Rotating arm (90 × 15 mm); **c)** Metal screw; **d)** Wire mold walker 1124L-1525 series locking nipple 1" – 1 1/4"; **e)** 12V DC motor; **f)** 10 mm PVC cup; **g)** Supportive stainless steel stand; **h)** Junction box (170 × 70 × 110 mm); **i)** Solar panel wire socket; **j)** DPDT On-Off switch rocker switch for solar panel; **k)** DPDT On-Off switch rocker switch for DC motor **l)** DC motor wire socket; **m)** Voltage regulator circuit board; **n)** 6V, 4.5-Ah rechargeable battery; **o)** Supportive ground stand; **p)** Solar panel (300 × 350 × 21 mm), module type: SEPL010P, Peak power voltage: 18 V, Peak power current: 0.56 A. Installation of spore trap at 0.5 and 2.5 m height from ground level in grapevine field at Onappalayam village of Coimbatore district **(B)**.

Table 5
Temporal variation in qPCR derived spore counts from DNA extracted from impaction spore traps during Rabi, 2024-25 and the respective spore counts/m^3 of air.

Date	Cq value		Spore count[a]		Spore count/m^3 of air	
	0.5 m	2.5 m	0.5 m	2.5 m	0.5 m	2.5 m
09-12-2024	28.31	28.42	6266.987429	5817.091329	12.845	11.92
16-12-2024	28.65	28.15	4978.045061	6984.215755	10.20	14.31
23-12-2024	24.32	28.54	93451.92146	5363.048997	191.55	10.99
30-12-2024	24.28	28.35	96018.06313	6099.498343	196.81	12.50
06-01-2025	18.78	24.25	3981071.706	97988.80708	8160.27	200.85
13-01-2025	13.54	24.21	138412869	100679.5293	283714.32	206.36
20-01-2025	13.85	24.21	5510315.563	91572.42303	11294.87	182.7
27-01-2025	18.30	28.35	185202.5913	6099.498343	379.622	12.5
03-01-2025	23.31	28.35	6266.987429	5817.091329	12.84	11.92

[a] Respective spore counts were calculated using formula as described by the Kunjeti et al. (2016).

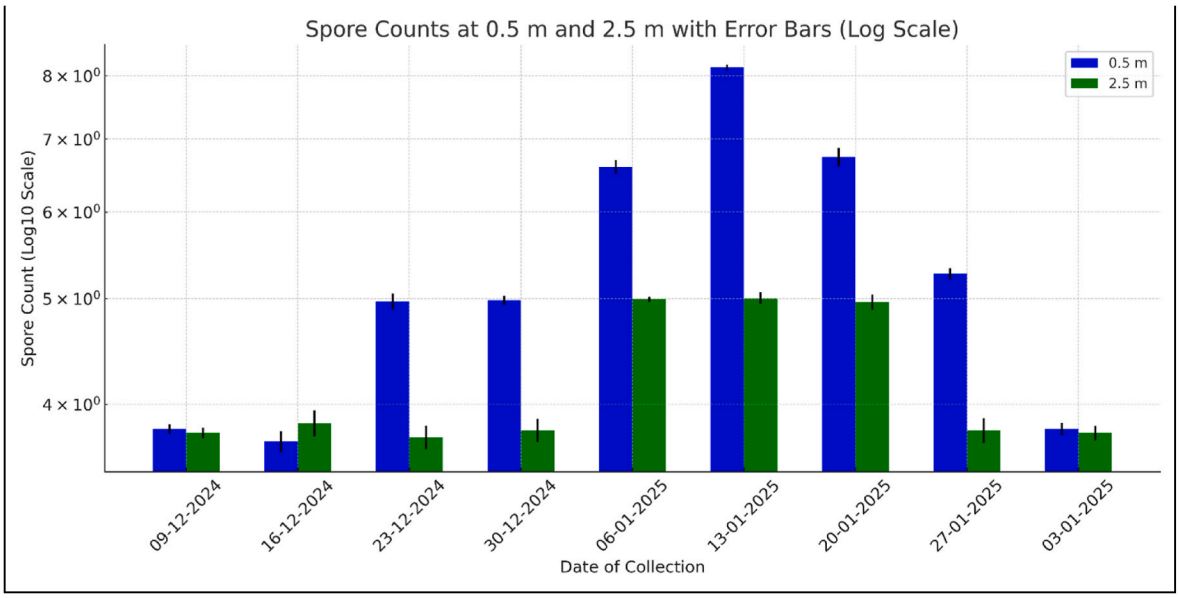


Fig. 8. The graphical representation of spore counts from respective spore trap along with date of sample collection. X – Date of collection of sampling rods from respective spore traps; Y- Sporangial counts in Log_{10} scale fashion. Each spore counts represent with respective error bar. Blue and green bar graph depict 0.5 m and 2.5 m spore count. *Respective spore counts were calculated using formula as described by the Kunjeti et al. (2016).

established a minimum sporangial threshold in air for initiating fungicide application. Notably, the spore traps positioned at lower heights within the grapevine canopy captured higher spore densities, underscoring the importance of trap height. These findings provide valuable insights into spore sampler deployment strategies and contribute to the advancement of effective disease monitoring systems for managing grapevine downy mildew.

CRediT authorship contribution statement

Govindan Muthukumar: Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Ayyanar Kamalakannan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Irudhayasamy Johnson:** Investigation, Formal analysis. **Pachamuthu Kamaraj:** Investigation, Formal analysis. **Iyyamperumal Muthuvel:** Investigation, Formal analysis. **Shanmugam Varanavasiappan:** Investigation, Formal analysis. **Thirumalai Angayarkanni:** Investigation, Formal analysis.

Declaration of competing interest

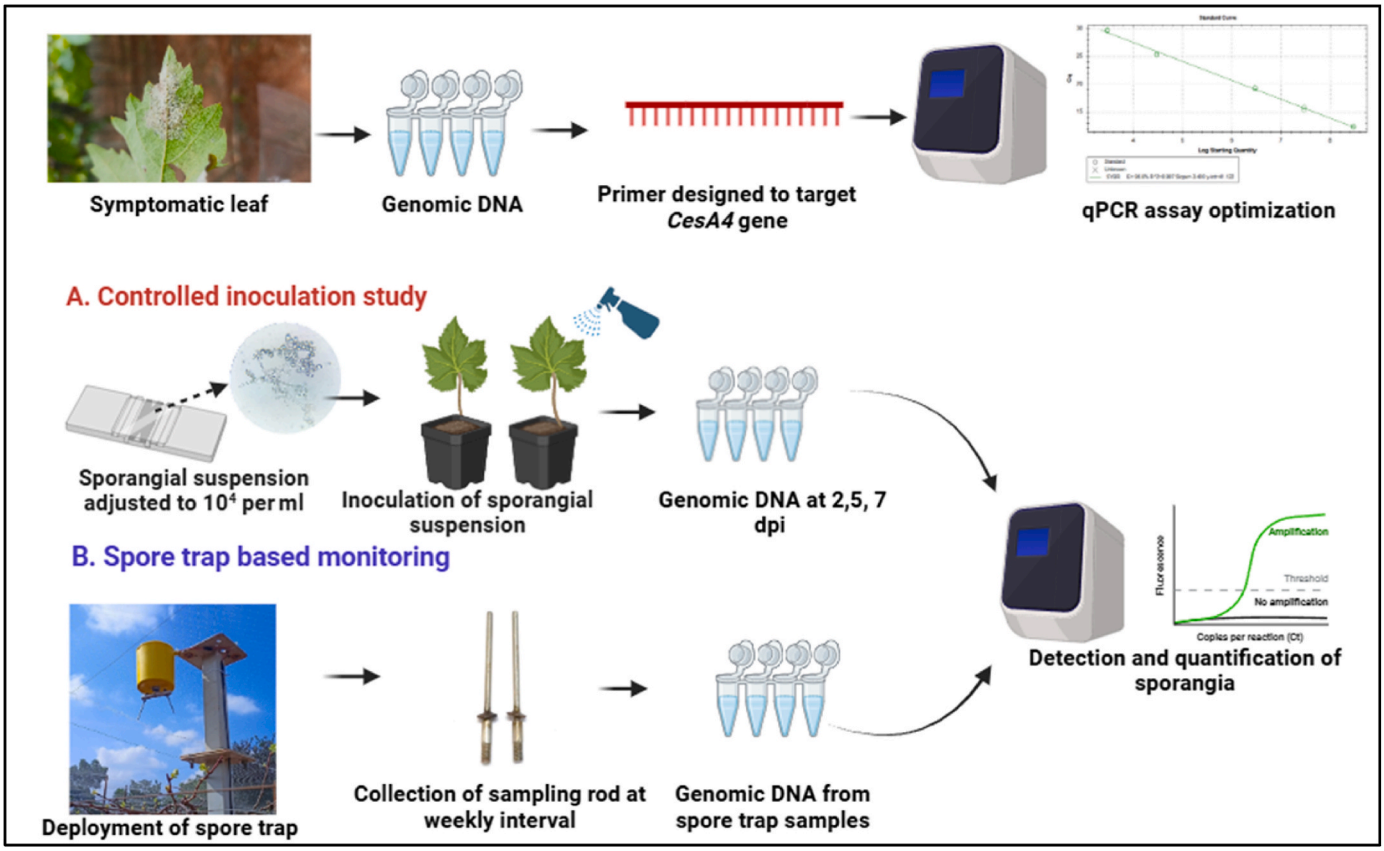
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pmpp.2025.102842.



Workflow for detection and quantification of *Plasmopara viticola* in asymptomatic leaves and spore trap samples.

Data availability

Included

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